Unfortunately, this does not improve the asymptotic execution time[7].

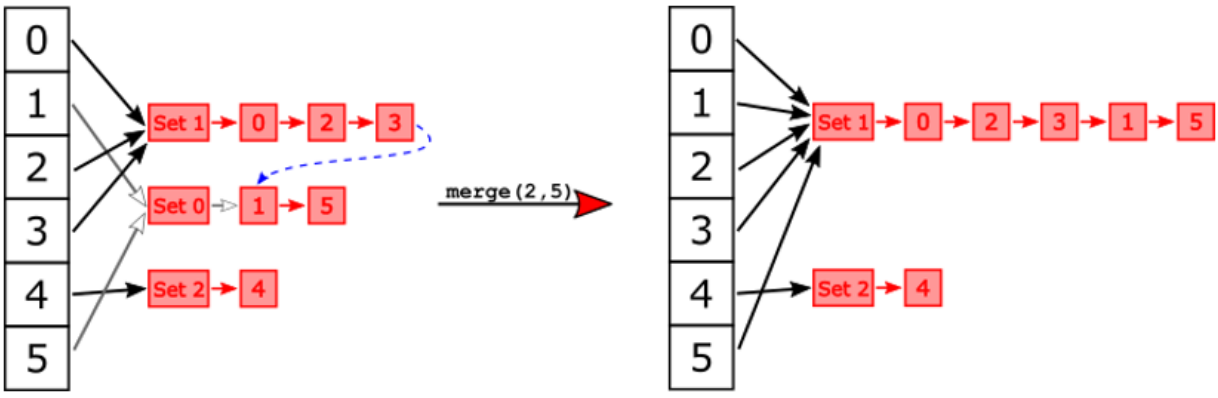


Figure 5.5 Merging two partitions. On the left: one of the two lists is appended to the other by adding a new edge tail-to-head, and removing links from the array to the second list. On the right it is shown the result after the merge, with all head-pointers for array elements belonging to the appended list (elements 1 and 5 in the example) updated.

Let's delve into code, though, to better explain how this works.

5.4.1 Implementing Naïve Solution

Let's start with the pseudo code for the constructor, and the class definition. All methods in the next sections will be assumed to be class methods for `DisjointSet`.

Listing 5.1 Naïve solution, constructor